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科目：概率論與隨機過程

1. Section 1.2 Problem 7.

Solution: The sample space is $\{(a_1, a_2, \dots, a_n) \in \{1, 2, 3, 4\}^n \mid 2 \mid \prod_{i=1}^{n-1} a_i \text{ and } 2 \mid a_n\}$.

2. Section 1.2 Problem 8.

Solution: Denote the possibility of winning player 1 is P_1 , winning player 2 is P_2 , winning player 3 is P_3 .

Without the loss of generality, $P_1 \geq P_2 \geq P_3$

Also, denote the possibility of winning the 1st round is x_1
 winning 2nd round is x_2
 winning 3rd round is x_3

Obvious, (x_1, x_2, x_3) is one of the rearrangement of (P_1, P_2, P_3)

$$P(\text{I win}) = x_2 x_1 + x_2 x_3 (1 - x_1) = x_2 (x_1 + x_3) - x_1 x_2 x_3 \\ = x_2 (x_1 + x_3) - P_1 P_2 P_3$$

$$x_2 (x_1 + x_3) \in \{P_1 (P_2 + P_3), P_2 (P_3 + P_1), P_3 (P_1 + P_2)\}$$

$$P_1 (P_2 + P_3) = P_1 P_2 + P_1 P_3 \geq P_1 P_2 + P_2 P_3 = P_2 (P_3 + P_1) \geq P_1 P_3 + P_2 P_3 = P_3 (P_1 + P_2)$$

Therefore, $P(\text{I win}) \leq P_1 (P_2 + P_3) - P_1 P_2 P_3$ (Equality holds when $x_2 = P_1$, $(x_1, x_3) = (P_2, P_3)$ or (P_3, P_2))

In other word, play the weakest opponent second, and that the order of playing the other two opponent doesn't matter

$$3. \text{Proof: } \forall x \in \left(\bigcup_i A_i\right)^c \Leftrightarrow x \notin \bigcup_i A_i \Leftrightarrow \forall i \in I, x \notin A_i \Leftrightarrow \forall x \in \bigcap_i A_i^c$$

$$\text{Therefore, } \left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c$$

$$\text{Meanwhile, } \left(\bigcap_i A_i\right)^c = \left(\left(\bigcup_i A_i^c\right)^c\right)^c = \bigcup_i A_i^c$$

Q.E.D.

4. Solution: Denote two ~~red~~ red cups are $C_{r,1}, C_{r,2}$ red saucers are $S_{r,1}, S_{r,2}$
white cups are $C_{w,1}, C_{w,2}$ white saucers are $S_{w,1}, S_{w,2}$
cups with stars on are $C_{s,1}, C_{s,2}$, saucers are $S_{s,1}, S_{s,2}$.

If two reds cups are placed on $S_{w,1}$ or $S_{w,2}$ saucers
 $C_{s,1}$'s cups must be placed on $S_{s,1}$ saucers
 $C_{s,2}$'s cups must be placed on $S_{s,2}$ saucers

Obviously, if we switch $S_{w,1}$ and $S_{s,1}$ saucers
we shall get the same result.

$$P(\text{Cr's are placed on } S_{s,1} \text{ or } S_{w,1}) = \frac{2 \times 2 \times 2 \times 2}{6!} = \frac{16}{720}$$

If $C_{r,1}$ is placed on one of the $S_{w,1}$, $C_{r,2}$ is placed on one of the $S_{s,1}$,
then we have to choose one of $C_{w,1}$ place it on the left of $S_{w,1}$
choose one of $C_{s,1}$ and place it on the left of $S_{s,1}$
The left of $C_{w,1}$ and $C_{s,1}$ can only be placed on $S_{r,1}$.

Obviously, we can get the same result if we switch $C_{r,1}$ and $C_{r,2}$.

$$\text{Therefore, } P(\text{one of } C_{r,1} \text{ is placed on one of } S_{w,1}, \text{ another one is placed on one of } S_{s,1}) \\ = \frac{2 \times 2 \times 2 \times 2 \times 2 \times 2}{6!} = \frac{64}{720}$$

Hence, the possibility that no cup is upon a saucer of the same pattern
is $\frac{16}{720} + \frac{64}{720} = \frac{1}{9}$.